ANNIHILATION OF EXT^1 PERSISTS UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-zero-ext-1-after-modding-out; the proof environment immediately following it is its own proof

Editorial note. The label lemma-zero-ext-1-after-modding-out names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the base change setup for pairs (A1, I1) to (A2, I2) of Noetherian adic rings (remark-base-change) and the base change B2 of a completion algebra B1 (equation-C-prime)
- the base change map for naive cotangent complexes (lemma-NL-base-change): a map NL_{B1/A1} tensor B2 B1 to NL_{B2/A2} inducing an isomorphism on H^0 and a surjection on H^{-1}
- the More on Algebra lemmas transferring vanishing/annihilation of Ext^1 under base change of a two-term complex and under a surjection on H^{-1} (more-algebra-lemma-two-term-base-change, more-algebra-lemma-base-change-property-ext-1-annihilated, more-algebra-lemma-surjection-property-ext-1-annihilated)

Source reference. Original equation 0AL4: sources/stacks-restricted/pinned-original-source.tex, source lines 326–326. Prerequisite pointer invoked at proof line(s) 1019; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GAG: sources/stacks-restricted/pinned-original-source.tex, source lines 669–669. Prerequisite pointer invoked at proof line(s) 1029; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0G9H: sources/stacks-chow/tags.txt, source lines 20302–20302. Prerequisite pointer invoked at proof line(s) 1036; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0G9J: sources/stacks-chow/tags.txt, source lines 20304–20304. Prerequisite pointer invoked at proof line(s) 1037; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0G9G: sources/stacks-chow/tags.txt, source lines 20301–20301. Prerequisite pointer invoked at proof line(s) 1035; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 06FQ: sources/stacks-restricted/pinned-original-source.tex, source lines 426–426. Prerequisite pointer invoked at proof line(s) 1018; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(A_1, I_1) \to (A_2, I_2)$ be as in Remark 06FQ with A_1 and A_2 Noetherian. Let B_1 be in (0AL4) for (A_1, I_1). Let B_2 be the base change of B_1. Let $f_1 \in B_1$ with image $f_2 \in B_2$. If $\mathrm{Ext}^1_{B_1}(NL^\wedge_{B_1/A_1}, N_1)$ is annihilated by f_1 for every B_1-module N_1, then $\mathrm{Ext}^1_{B_2}(NL^\wedge_{B_2/A_2}, N_2)$ is annihilated by f_2 for every B_2-module N_2.*

Proof. By Lemma 0GAG there is a map

$$NL_{B_1/A_1} \otimes_{B_2} B_1 \to NL_{B_2/A_2}$$

which induces and isomorphism on H^0 and a surjection on H^{-1}. Thus the result by More on Algebra, Lemmas 0G9G, 0G9H, and 0G9J the last two applied with the principal ideals $(f_1) \subset B_1$ and $(f_2) \subset B_2$. □